

Lecture 3 Homework

<http://thegrantlab.org/>

Dr. Barry Grant

Your Name: Derek Zhang

UCSD Email: Dez004@ucsd.edu

PID: A17819201

To complete this homework you must be working on your own copy in **Google Docs**. Once you have filled in your answers to **Q1-Q4** in the space provided click **File > Download > PDF document** and upload to **gradescope** (link can be found on the class website).

Q1. [4pts] Consider the following multiple alignment of Transcription Factor Binding site DNA sequences

	1	2	3	4	5
Sequence 1	-	G	A	G	C
Sequence 2	C	T	A	G	A
Sequence 3	C	G	A	-	A
Sequence 4	A	G	C	G	A

Give the average profile (frequency matrix) of the above alignment by filling out the table below. The first position of the first column (i.e. position in the alignment) has been done for you, now complete the rest. You will use this table for answering questions 2 and 3 below.

	1	2	3	4	5
A	0.25	0	0.75	0	0.75
C	0.5	0	0.25	0	0.25
T	0	0.25	0	0	0
G	0	0.75	0	0.75	0
-	0.25	0	0	0.25	0

Q2. [2pts] What is the highest scoring sequence match to your profile above (question 1) and what is its score?

Sequence: CGAGA

Score: 3.5

Q3. [2pts] Using your completed profile table above (from question 1) score the following two sequences (**S1** and **S2**):

S1. CTGGC: 1.75

S2. AGAGA: 3.25

Q4. [2pts] Following the heuristic threshold for a positive match proposed in Harbison et al. [Nature (2004) 431:99-104.] namely using the threshold for a positive match = $60\% \times \text{Max Score}$
Are either of the two sequences in question 3 potential transcription factor binding sites? If so, why?

The threshold for a positive match being $0.6 \times \text{max score}$ yields a threshold of 2.1. This means that only sequence 2 in question 3 is a positive match as its score is 3.25 and is greater than the threshold of 2.1.